



## PG Diploma in Embedded and Automotive Systems

Duration – 5 months

Eligibility – B.E, B. Tech, M.E, M. Tech.

### Modules

#### Generic Modules

- Electronics and Embedded Hardware Familiarization
- System Development Life Cycle
- Verification and validation

#### Core Programming

- Programming in C and C++ following MISRA
- Linux Commands and Shell Scripting
- ARM CORTEX M4 Programming using Embedded C
- System Programming Using C
- Basic Python Programming

#### Embedded Specialization

- ARM Bare Metal Programming using Embedded C
- Linux Device Driver Programming
- Embedded OS (RTOS) Programming
- DevOps for Embedded Software Development

#### Automotive Specialization

- Automotive Protocols & Autosar Architecture
- MBD for Autosar Model Deployment using Matlab & Simulink

### Project stream:

- Application development based on Data Structure (Eg: Flood fund releasing data, cyber management systems, Bank management system, contact management system)
- Concurrent server to replicate a chat application using socket programming.
- Embedded Project development using wired/wireless technology such as GPS, GSM, BLE, Wi-Fi, and sensors
- Project Based on CAN or Lin Protocol
- Parallel port Driver.

### Platform:

- Ubuntu (Linux OS, with gcc compiler)
- Lpc2129 development board
- Keil Micro Vision, Flash Magic
- STM32F411RE Nucleo64 Board
- STM CUBEIDE

## Generic Modules

### Electronics and Embedded Hardware Familiarization – 10 Days

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|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------|
| Analog Electronics: Passive and Active components                                   | Circuit analysis using KCL and KVL                                      | Diode, Transistor and Op-amp Circuits          |
| Digital Electronics: Combinational circuits design: Adders, Mux, Encoder, Decoder   | Sequential circuits design: Flipflops, Registers, Counters              | Microprocessors, Microcontrollers,             |
| Basic Embedded System Architecture                                                  | Standard Interfaces                                                     | Understanding schematics/datasheet             |
| Fundamentals of Booting for Embedded Processors: Host and Target Development Setups | Electrostatic Discharge Essentials: Causes of ESD and Prevention of ESD | Techniques to improve embedded system security |

### System Development life Cycle – 05 Days

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|----------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| SDLC Overview and its Phases                                         | Embedded System Design Cycle - Hardware and Software Aspects                             | SDLC Models: V model, Water Fall model                                            |
| Agile - an iterative and responsive software development methodology | Architecting and Designing the System, Reviews, Languages and Static Code Analysis (SCA) | Measurement and Metrics, Development and Operations, Overview of Embedded Testing |



| Verification & Validation - 05 days                                                                  |                                                                                                            |                                                                                                                             |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Fundamentals of testing: Testing principles, test process                                            | Testing Throughout the SDLC: Test levels, Test types                                                       | Static Techniques: Static testing basics, review process                                                                    |
| Test techniques: Black -box, White-box, Experience based test techniques                             | Test Design techniques for Embedded System                                                                 | Test automation, Test Monitoring & Control                                                                                  |
| Core Programming                                                                                     |                                                                                                            |                                                                                                                             |
| Programming in C and C++ following MISRA - 15 days                                                   |                                                                                                            |                                                                                                                             |
| Introduction to C: Simple C program structure, Literals, constants, variables and data types         | Operators with precedence and associativity                                                                | Control flow statements with Examples                                                                                       |
| Modular Programming using functions                                                                  | Working with multiple files                                                                                | Storage Class Specifiers                                                                                                    |
| Arrays and Strings                                                                                   | Preprocessor directives                                                                                    | Pointer declaration, initialization, pointer arithmetic                                                                     |
| Stack Implementation using array                                                                     | Queue Implementation using array                                                                           | Introduction to C++, Structure of a C++ program                                                                             |
| Function Overloading in C++, Default argument functions                                              | Function templates                                                                                         | Classes and objects                                                                                                         |
| Constructors and Destructors                                                                         | Inheritance                                                                                                | Runtime Polymorphism                                                                                                        |
| Linux Commands & Shell Scripting – 05 days                                                           |                                                                                                            |                                                                                                                             |
| Introduction to the Linux operating system                                                           | Linux Commands: Basic commands, process commands, file commands                                            | Introduction to shell scripting                                                                                             |
| Shell scripting Input and output                                                                     | Arithmetic Expression                                                                                      | Decision making and Loop Constructs                                                                                         |
| ARM Cortex- M4 Programming using Embedded C – 05 days                                                |                                                                                                            |                                                                                                                             |
| ARM Cortex M4 CPU Architecture                                                                       | STM32F411 RE Microcontroller Features, On-chip peripherals                                                 | GPIO Registers and GPIO Programming on STM32                                                                                |
| UART Registers and UART Programming on STM32                                                         | Timer Register Configuration and Programming on STM32                                                      | VIC registers and interrupt Programming on STM32                                                                            |
| System programming using C – 10 days                                                                 |                                                                                                            |                                                                                                                             |
| Building an executable: MAP file basics, Building libraries                                          | Debugging & Tracing with GDB: Debug options, disassembly of c code                                         | Memory Management: Memory Architecture, allocation, profiling                                                               |
| Advanced Data Types: Variable length Arrays, typedef, typecasting                                    | Structures and Unions: Copying Structure, Nested Structures, Passing a Structure to a Function, Endianness | various pointers, Handling arrays, structures, functions with pointers                                                      |
| Working with registers: Bit level operations, Handling special registers. const, volatile qualifiers | Context management: Code/Data/BSS/Stack/Heap segments, Execution context, Interrupt Context                | Linux fundamentals: Linux Kernel basics, Linux driver framework and filesystem, Networking sockets                          |
| C Library functions: Useful primitives, System Calls, OOPS Concept                                   | Inter process communications: Handling a Signal using sigaction, Socket programming                        | Coding practices: Coding guidelines, comments, conditional compilation<br>Secure and Safe Coding<br>Developing optimal code |
| Basic Python Programming – 05 days                                                                   |                                                                                                            |                                                                                                                             |



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|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Introduction to Python                                                        | Python Data types , input and output statements                                                                         | Control Statements : conditional and iterative statements                                                      |
| Python Function syntax and examples                                           | List and tuple examples                                                                                                 | Set and dictionary examples                                                                                    |
| Lambda , filter and map functions                                             | Exception handling                                                                                                      | Working with multiple files                                                                                    |
| <b>Embedded Specialization</b>                                                |                                                                                                                         |                                                                                                                |
| <b>ARM Bare Metal Programming using Embedded C – 20 days</b>                  |                                                                                                                         |                                                                                                                |
| ARM7TDMI Architecture and features                                            | LPC2129 Features                                                                                                        | GPIO programming with LED, Switch and Buzzer                                                                   |
| 16 X 2 LCD programming                                                        | 4 X 4 KEYPAD programming                                                                                                | ADC programming                                                                                                |
| Timer programming                                                             | PWM programming                                                                                                         | RTC and WDT programming                                                                                        |
| PLL programming                                                               | VIC (Interrupt) programming                                                                                             | UART programming                                                                                               |
| SPI programming                                                               | I2C Programming                                                                                                         |                                                                                                                |
| <b>Linux Device Driver With Beagle bone –10 days</b>                          |                                                                                                                         |                                                                                                                |
| Introduction to the device driver<br>Creating a kernel module                 | uploading and removing the module from the kernel space                                                                 | Makefile                                                                                                       |
| Character driver module                                                       | Block driver module                                                                                                     | USB Device driver                                                                                              |
| Introduction to network driver                                                | AM335x Functional block and overview                                                                                    | U-BOOT Mechanism & BEAGLE BONE eMMC                                                                            |
| <b>Embedded OS (RTOS) Programming – 10 days</b>                               |                                                                                                                         |                                                                                                                |
| Overview of FreeRTOS: Features of freeRTOS, FreeRTOS source code organization | RTOS Concepts : Hard real time vs soft real time , Multi-threading/ Multi-tasking / Concurrent execution                | Scheduling and Context switching                                                                               |
| Memory management: Heap vs Stack memory, program memory vs data memory        | freeRTOS Heap Memory Management ,different memory allocation schemes freeRTOS Heap Utility Functions, Optimizing memory | Concept of freeRTOS Tasks<br>freeRTOS Tasks APIs, Creating Tasks, Task Priorities , Task State Transitions     |
| Scheduler : Scheduler Algorithms, Tick Interrupt, Idle task                   | Inter task Communication and synchronization: freeRTOS Queue APIs<br>Data storage for Queue                             | Blocking read, write<br>Receiving data from multiple queues<br>Mailbox (using queue)                           |
| Interrupt Management<br>Events and ISRs ,Tasks vs ISRs                        | Semaphores<br>Concept of semaphores ,Binary Semaphores, Counting semaphores                                             | Resource Management :<br>Shared resources<br>Mutual Exclusion ,Critical Section                                |
| Mutex<br>Deadlocks, starvation , Priority inversion , Priority Inheritance    | Event Groups :<br>Event Groups for Multiple Task Synchronization                                                        | Task Notifications<br>Tasks communication objects vs direct task notifications ,Benefits of task notifications |
| <b>DevOps for Embedded Software Development – 05 days</b>                     |                                                                                                                         |                                                                                                                |
| Introduction to DevOps                                                        | Basic of Virtualization                                                                                                 | Version Control System                                                                                         |
| Git and GitHub                                                                | Building and Packaging                                                                                                  | Continuous Integration and Delivery with Jenkins                                                               |
| <b>Automotive Specialization</b>                                              |                                                                                                                         |                                                                                                                |



## Automotive Protocols & Autosar Architecture – 10 days

|                                              |                                         |                                                 |
|----------------------------------------------|-----------------------------------------|-------------------------------------------------|
| Automotive Systems Overview                  | CAN Architecture                        | CAN Framing                                     |
| CAN arbitration                              | CAN programming                         | CAN Frame Analysis Using CAN-BUSMASTER Analyzer |
| LIN Bus : Working Principle and Applications | LIN Bus v/s CAN Bus<br>LIN Frame Format | AUTOSAR layer Architecture                      |
| Diagnostics: On Board Diagnostics (OBD)      | Unified Diagnostic Services (UDS)       | Firmware On the Air (FOTA)                      |

## MBD for Autosar Model Deployment using Matlab & Simulink – 10 days

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|----------------------------------------------------------------------------|-------------------------------------|---------------------------------------------------------------|
| Introduction to MATLAB- Data Types, Addressing Techniques, linear Equation | Creating and manipulating matrices- | Data Visualization-2D and 3D plots, Digital image processing, |
| Programming in MATLAB                                                      | Graphical User Interface            | Introduction to Model-Based Design                            |
| Logic Driven Modeling                                                      | Finite state machines               | Introduction to Simscape                                      |